SimpleTracking Instructions

SimpleTracking Version 2.0

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This is the **first step** of chemotaxis analysis in I2Tumbles. `SimpleTracking.m` converts a series of `.tif` images into a `.mat` file and an excel file that contains X and Y coordinates of the bacterial movements. If you can't find the right setting from `SimpleTracking.m`, we recommend using TrackingGUI and use the resulting `.mat` file for the tumble analysis (`CellTrackAnalysis.m`).

Step 0: Adjusting Videos to work in SimpleTracking

Before running SimpleTracking, bacterial movement videos should be converted to `.tif` image files. See Kailie's tutorial here to learn how to convert videos to image files and parameter settings.

Step 1: SimpleTracking Parameters

1. Open MATLAB and open `SimpleTracking.m`
2. Adjust Main parameters. The most important parameters are `numFrames` and `secondVideo`. Make sure those two parameters are correct.

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
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%% MAIN PARAMETERS

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
bpfiltsize=         20;                         % bpfiltsize        [A]
hthreshopt=         2;                          % Threshold Option  [B] 2. Std. Thresh.
defaultthresh2=     4.0;                        % thr (>1)          [C]

stdevMinThreshold=  2;                          % Cull: Std. Min. Threshold
numFrames=          200;                        % Cull: # of Video Image Frames
secondVideo=        10;                         % Cull: Length of Video (sec)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```
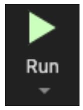

3. Adjust other parameters if necessary. All the parameters are pre-set to the setting that is frequently used in the lab. See Kailie's tutorial or TrackingGUI manual if you are unsure which parameters you should use.

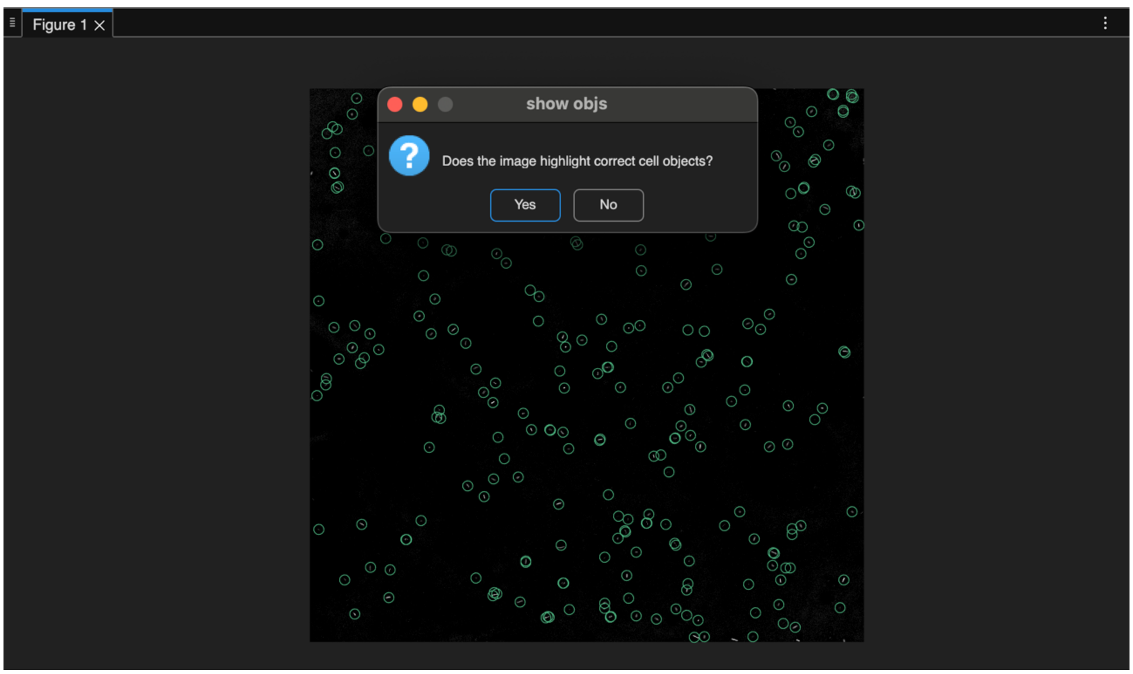
Step 2: Select Images

1. Click Run 

2. The script will prompt you to select the image files. Browse to the folder with the images
3. Select the first image (format should be `.tif`)
4. Select the last image

Step 3: Confirm Objects

1. The first image of the video will pop up, with green circles on each bacterium object to show that the software will track them.
2. Click "Yes" if the objects look correct. Otherwise, click "No" and the software will stop. Adjust parameters and re-run if incorrect.



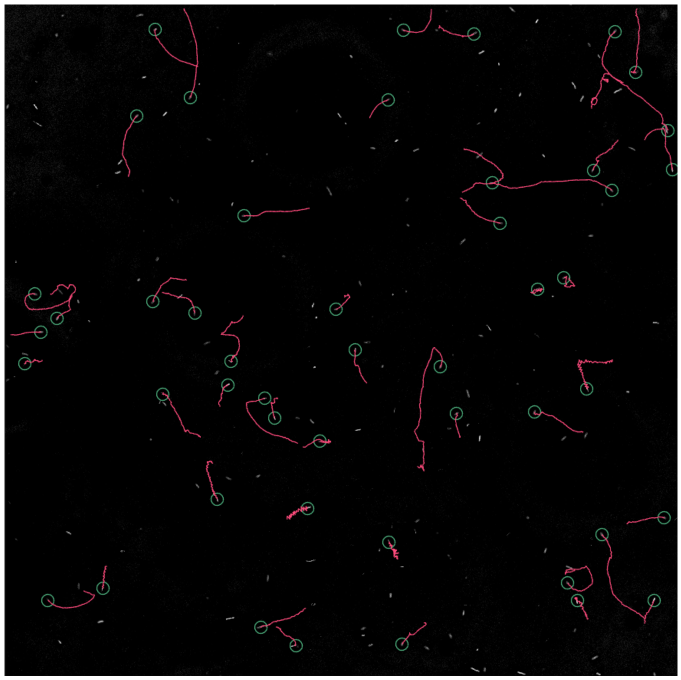
Step 4: Track, Link and Cull Objects

1. After confirming, the software will proceed with tracking, Linking and Culling all objects. You can see the progress with the progress bar.
2. If parameters aren't changed, the software will cull twice (`Stdev`, `TrackLength`)
3. You'll be able to see the number of tracks after culling on the Command Window. Copy the number if needed. In the screenshot, 6036 is # tracks before the cull and 284 is after.

```
>> SimpleTracking
Found 6036 unique tracks.
More than 10 tracks found; not displaying details.
Number of Tracks After Cull:
   284
```

Step 5: Save Objects

1. Next, another popup will show with the object tracks after the cull with red lines. Confirm that the tracks look correct.



2. SimpleTracking saves the output and automatically runs Kailie's additional scripts and `migration_v2.m` afterwards.
3. Outputs
    a. The outputs will be overwritten if you run this script again – save the outputs in a different location or change the name of them if necessary.
    b. `SimpleTrackingoutput.mat`
        i. This `.mat` file is a collection of variables and metadata that will be used in `CellTrackAnalysis.m` (Step 2 of I2Tumbles).
    c. `Tracked_positions.xlsx`
        i. This excel file contains the X and Y coordinates of each bacteria and couple of metadata.

References

- TrackingGUI GitHub: https://github.com/rplab/TrackingGUI_and_Localization_Public
- ImageJ and TrackingGUI instructions by Kailie Franco: https://github.com/clara-h-shin/I2Tumbles/blob/main/swim_tracking_tutorial_by_kailie.docx